Jiaheng Li

Full Stack Software Engineer

odell.lijiaheng@gmail.com
Portfolio · GitHub · Auckland, NZ
Open Work Visa to 25 Feb 2029 · No sponsorship required
Available immediately · Open to relocate

At a Glance

Experience: 3 years commercial development

Data: PostgreSQL · Oracle · MongoDB

Systems: Security · Distributed · Blockchain · Networks · FFmpeg · Stripe · OAuth 2.0

Languages: Go · Python · Java · C# · C/C++ · JS · TS

Backend: Gin · Django · Flask · FastAPI · REST API

Frontend: React · Node.js

Professional Experience

Skyline Consulting Engineers Limited

Oct 2024 - Dec 2025

Part-time Full-stack Developer · Auckland

Turned structural engineers' calculation workflows into C#/.NET, React/TypeScript and PostgreSQL software, validating results against analytical benchmarks. Integrated Stripe subscription billing and Google sign-in using OAuth 2.0 and ASP.NET Core Identity.

Yonyou Network Technology Co., Ltd.

Mar 2022 - Dec 2023

Java Backend Developer · Zhengzhou, China

Developed Java/Spring REST APIs and accounting forms for a multi-tenant financial SaaS ERP. Worked with Oracle, PostgreSQL and MyBatis, and built cross-module login and Kafka message processing for workloads above 100,000 records.

Projects

Go Kademlia DHT
Go · DHT · P2P
Concurrent networking

Implemented four Kademlia RPCs in Go, with XOR-distance routing and concurrent queries. Integrated the DHT with a torrent-download session; documented peer discovery, packet captures and a successful torrent download in my undergraduate thesis.

Blockchain PKI
Blockchain · PKI
Network security

Explored lightweight blockchain-based certificate issuance, validation and revocation, including smart-contract concepts; produced a postgraduate research report.

P2P Lockstep Kit
P2P protocol · WebRTC
WebSocket · State sync
TypeScript

Built a browser-game toolkit with separate network, session and UI packages. WebSocket signalling connects peers over WebRTC; sessions handle ordered moves, Ready/Start, undo/restart and reconnect synchronisation. Web Components and a game-plugin interface separate rules from networking. Powers four live games, including multiplayer Mahjong, with an AI Skill for building new games.

Live demos: Mahjong · Gomoku · Chess · Xiangqi
npm packages: Network · Session · UI · AI build Skill

NX-Cast
DLNA · AirPlay · IPTV
FFmpeg · libmpv

Turned a Nintendo Switch into a media receiver supporting DLNA, IPTV and AirPlay 2. Cross-compiled and integrated FFmpeg/libmpv, using event loops, multithreading and state machines to manage playback within embedded memory and CPU constraints.

Total-Order Multicast
C# · Distributed protocols
TCP · Logical clocks

Implemented and tested consistent message-delivery ordering across five processes under simulated network delays as a distributed-computing coursework project.

StructLedger
.NET · React · PostgreSQL
Stripe · OAuth 2.0

Released a beam-calculation web application with Stripe Checkout, a customer billing portal and signed webhook validation to synchronise subscription access. Added Google sign-in and CI/CD deployment to Railway and Cloudflare.

Education

Master of Information Technology · University of Auckland · Sep 2026

Bachelor of Network Engineering · Henan University · Jun 2022

Former employers are willing to provide references.